Calcurion Maths Worksheets

A-Level Further Mathematics

Topic: Core Pure 2 – Applications of Integration

Medium & Hard • 25 Questions

Scan for free resources

Section A – Medium (Q1–15)

1. State the formula for volume of revolution about the x-axis. **[2]**

2. State the formula for volume of revolution about the y-axis. **[2]**

3. State the formula for arc length of $y = f(x)$. **[2]**

4. Find the volume when $y = x$ from 0 to 1 is rotated about the x-axis. **[3]**

5. Find the volume when $y = e^{-x}$ from 0 to 1 is rotated about the x-axis. **[4]**

6. Find the arc length of $y = \ln x$ from $x = 1$ to $x = e$. **[5]**

7. Find the volume when the area under $y = \frac{1}{x}$ from 1 to 2 is rotated about the x-axis. **[4]**

8. Find the centroid $\bar{x}$ of the region under $y = x^2$ from 0 to 1. **[4]**

9. Find the centroid $\bar{y}$ of the region under $y = x^2$ from 0 to 1. **[4]**

10. State Pappus's theorem relating centroid and volume of revolution. **[3]**

11. Find the surface area generated when $y = x$ from 0 to 1 is rotated about the x-axis. **[4]**

12. Find the volume of a sphere of radius a by integration. **[4]**

13. Find the arc length of the parametric curve $x = t^2$, $y = t^3$ from $t = 0$ to $t = 1$. **[5]**

14. Find the volume when $y = \sin x$ $(0$ to $\pi)$ is rotated about the x-axis. **[4]**

15. Find $\bar{x}$ for the region under $y = \sqrt{x}$ from 0 to 4. **[4]**

Section B – Hard (Q16–25)

16. Find the volume of the solid formed by rotating the region bounded by $y = x^2$ and $y = x$ about the x-axis. **[5]**

17. Find the surface area of a sphere of radius a by integration. **[5]**

18. Find the arc length of $y = \cosh x$ from $x = 0$ to $x = 1$. **[4]**

19. Find the volume generated when the area under $y = \frac{1}{x}$ from 1 to ∞ is rotated about the x-axis (Gabriel's horn) and comment on the surface area. **[6]**

20. Use the parametric equations of a cycloid to find one arch arc length. **[6]**

21. Find the coordinates of the centroid of a semicircular lamina of radius a. **[5]**

22. Find the volume when the region between $y = x^2$ and $y = 2x$ is rotated about the y-axis. **[5]**

23. Find the arc length of $r = a(1 + \cos\theta)$ from $\theta = 0$ to $\theta = \pi$. **[5]**

24. Find the surface area generated by rotating $y = \sqrt{x}$ from 1 to 4 about the x-axis. **[5]**

25. Use Pappus to find the volume of a torus with major radius R and minor radius r. **[4]**

End of Worksheet • *Show all working clearly.*